MILITARY SPECIFICATION

# ANTENNAS AT 500( * )/AP AND AT-929( * )/AP; AND ANTENNA BASE AB 117( * )/AP

This specification has been approved by the Bureau of Aeronautics, Department of the Navy.

## 1. SCOPE

1 1 Scope - This specification establishes the requirements for Antennas AT-500( * )/AP (300 to 1000 mc/s); AT-929( * )/AP (80 to 320 mc/s); and Antenna Base AB-117( * )/AP which supports the two antennas interchangeably. The antennas are of the streamlined blade type for use with airborne electronic equipment.

1. 2 Classification - The equipment covered by this specification shall be of the following types

| Item | Type Designation | Max. Allow. Size (Inches) | | | Max. Allow. Wt. (Lbs.) |
|---|---|---|---|---|---|
| Antenna | AT-500( * )/AP | 14-3/4 | 6-1/2 | 3-1/2 | 3.5 |
| Antenna | AT-929( * )/AP | 35-1/2 | 6-1/2 | 3-1/2 | 7.0 |
| Antenna Base | AB-117( * )/AP | 11-1/4 | 4-1/2 | 3 | 3.2 |

See 6.3

## 2. APPLICABLE DOCUMENTS

2. 1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |

Military

| | |
|---|---|
| MIL-C-71 | Connectors, "N" for Radio Frequency Cables |
| MIL-T-152 | Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment |
| MIL-V-173 | Varnish, Moisture-and Fungus-Resistant, For the Treatment of Communications, Electronic and Associated Electrical Equipment |

FSC 5985



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:27Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification For |
| MIL-T-5422 | Testing, Environmental, Aircraft Electronic Equipment |
| MIL-C-7439 | Coating System, Elastomeric, Rain Erosion Resistant and Rain Erosion Resistant with Anti-Static Treatment, For Exterior Aircraft and Missile Plastic Parts |
| MIL-N-18307 | Nomenclature and Nameplates for Airborne Electronic and Associated Equipment |
| MIL-P-17555 | Preparation for Delivery of Electronic Equipment, Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts |

DRAWINGS

Bureau of Aeronautics

| | |
|---|---|
| 60A1D40-21 | AT-500/AP Antenna, Assembly |
| 60A1D40-22 | AT-929/AP Antenna, Assembly |
| 58A1C40-16 | AB-117/AP Antenna Base, Assembly |
| 58A1D40-17 | AB-117/AP Antenna Base, Detail Drawing |

Bureau of Ships

| | |
|---|---|
| RE49F290 | UG-98(*)/U Radio Frequency Receptacle |
| REB49338 | UG-100(*)/U Radio Frequency Plug |

2.2 Availability of Documents

(1) When requesting applicable documents give both the title and number All requests shall be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, (CDS), Philadelphia 20, Pennsylvania.

2.3 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over all referenced documents.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:27Z
Check the source to verify that this is the current version before use.

(2) This Specification - This specification shall have precedence over all applicable subsidiary documents. Deviations from this specification, or from subsidiary documents where applicable, may be made only upon written approval.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

## 3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the characteristics of standard MIL parts and materials are such their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specification.

(3) When a nonstandard part is used, the meeting of general material requirements such as nonflammable, fungus resistant, nontoxic, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the Bureau of Aeronautics.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400, except as follows.

(1) Mechanical parts shall be submitted as nonstandard parts only when the material requirements of Specification MIL-E-5400 are not met.

(2) Nonstandard parts of an electrical nature as defined in Specification MIL-E-5400 shall be approved before the preproduction model of the equipment is submitted for tests. The request for approval shall be forwarded to the Bureau of Aeronautics before the part is purchased or fabricated. In addition to or in lieu of the Nonstandard Part Data Sheet, the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:27Z
Check the source to verify that this is the current version before use.

3.1.2 Cadmium Plating - Cadmium plating in accordance with Specification QQ-P-416 shall be used to insure electrical contact between brass and aluminum alloy metal.

3.1.3 Soldering - Soldering shall be in accordance with the requirements of Specification MIL-E-5400. Cleaning flux, such as sal amoniac, shall not be used. Alcohol may be used for cleaning after tinning and soldering. Mechanical loads shall not be imposed on soldered connections.

3.1.4 Dielectric Material - The dielectric material or materials used between the stub and the sleeve of the antenna shall be chosen so that the antenna will best fulfill the requirements of this specification. If impregnated glass cloth is used in the construction of the antenna it shall be Corning Glass works type ECC-162-11 (or finer mesh) or an approved equivalent and shall be impregnated with a suitable polyester resin having a dielectric constant of 2.75 ± .25 and a dissipation factor of no greater than .014 at 200 megacycles per second.

3.2 Design and Construction:

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Construction - The antenna shall have the airfoil mounting dimensions, and general features of construction as shown on Bureau of Aeronautics Drawings listed in 2.1 The bottom portion of the antenna, which mates with the outer surface of the aircraft skin, shall be so designed as to permit essentially a high pressure electrical contact between the antenna base and the base surface of the antenna, and at the same time allow for the use of small moisture sealing gasket within the antenna base, without interfering with the electrical contact with the skin. Cementing of dielectric material, although desirable for sealing purposes shall not be relied upon to secure the dielectric portion of the antenna to its upper and lower metal portions. Other provisions, such as flush rivets or bolts used with grommets or pressure plates, shall be employed to insure adequate structural strength of the antenna. The external dielectric surface shall be coated with a nonconducting, low loss dielectric material to prevent absorption of moisture or the formation of a film of moisture on the surface. In addition a Class I rain erosion coating shall be applied to the antenna in accordance with the requirements of Specification MIL-C-7439.

3.2.3 Mechanical Strength - The antenna shall withstand a uniformly distributed lateral ultimate design load of 9.5 pounds per square inch, when properly mounted in an AB-117( * )/AP Antenna Base The Antenna Base AB-117( * )/AP shall withstand the above load with an AT-929( * )/AP properly installed and the antenna base mounted in a jig which will properly support it using the standard mounting (four bolts) specified in Drawing 58A1C40-16.

3.2.4 Mating Surfaces - Electrically conducting mating surfaces shall be clean metal surfaces, free from grease, paint lacquer, anodizing or similar resistive film so as to insure negligible radio frequency impedance

3.2.5 R-F Connectors - The R-F connection included as part of these antennas shall be of the LN type and shall have the mating surfaces of a UG-98( * )/U in accordance with Bureau of Ships drawing RE49F290. It shall accommodate connector UG-100( * )/U in accordance with Bureau of Ships drawing REB49338.

3.2.6 Safetying - All internally and externally threaded parts shall be positively safetied by the use of lock washers, lock nuts, or other suitable methods.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:27Z
Check the source to verify that this is the current version before use.

3.2.7 Weight - The antennas shall be of the lightest practical weight consistent with other requirements specified herein and in no event shall the installed weight of the antenna exceed that specified in 1.2. Installed weight shall include the complete antenna assembly with nameplate and receptacle, but shall not include cabling, external to the antenna, plugs, mounting nuts or bolts, or any mounting bracket or fastening device which is not an integral part of the antenna assembly. The installed weight of the Antenna Base AB-117( * )/AP shall include the base casting, cover plate, gasket, screws and nuts or other fastening devices for mounting antennas or cover plate and the four mounting bolts with nuts and spacing sleeves as shown on Drawings 58A1C40-16 and 58A1D40-17.

3.2.8 Tropicalization - Antennas covered by this specification shall be tropicalized in accordance with Specification MIL-T-152 and MIL-V-173.

3.2.9 Collection of Liquids - The antenna shall be so constructed that when mounted vertically either on the top or bottom surface of an aircraft structure no water or other liquids can collect in any portion of the antenna that could ultimately result in failure or deterioration of performance of the antenna or rupture due to freezing.

3.2.10 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307. See 6.3.

3.2.11 Interchangeability - The equipment shall meet the interchangeability requirements as specified in Specification MIL-E-5400. Unless otherwise specified, interchangeability shall be measured against a model supplied by the Bureau of Aeronautics.

3.2.12 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.13 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

| | |
|---|---|
| Temperature | Room ambient (30° ± 10°C) |
| Altitude | Normal ground |
| Vibration | None |
| Humidity | Room ambient up to 90% relative humidity |
| Input power voltage | 115 + 1.0 V AC and/or 27.5 + 0.5 V DC |

3.2.14 Service Conditions - The equipment shall operate under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class I equipment.

3.3 Performance.

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Radiation Characteristics - The antenna shall be so constructed that when mounted in the center of a large horizontal ground plane it shall provide vertically polarized radiation of essentially uniform intensity over 360° of azimuth. The antenna shall satisfy the performance requirements which are outlined in Section 4 when subjected to the following tests:



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:27Z
Check the source to verify that this is the current version before use.

(1) Mechanical Stress Test
(2) Insulation Test
(3) Radiation Test
(4) Voltage Standing Wave Ratio Test

3.4 Detail Requirements·

3.4.1 General - Antennas AT-500( * )/AP, and AT-929( * )/AP, and Antenna Base AB-117( * )/AP shall conform to the general configuration shown on the applicable drawings listed in 2.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - The antennas shall be subjected to the following tests to determine compliance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on 3 each of the antennas representative of the production antennas to be supplied under the contract. Preproduction tests shall consist of the following·

(1) Contractor's Deminstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.4. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the antennas to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the Bureau of Aeronautics

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the Bureau of Aeronautics, the antennas, plainly identified by securely attached durable tags marked with the information specified herein, shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests deemed necessary to determine compliance with all applicable design and performance requirements·

Sample for Preproduction Test
Antenna Type
Name of Manufacturer
Submitted by (name) (date) for qualification in accordance with requirements of Specification MIL-A-22212 under authorization (reference authorizing letter)

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-T-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:27Z
Check the source to verify that this is the current version before use.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the antennas meet all the requirements of this specification and the contract and shall consist of at least the following tests conducted in the order stated. The antenna base shall be subjected to the first two tests.

(1) Environmental Tests

(2) Mechanical Stress Test

(3) Insulation Test

(4) Voltage Standing Wave Ratio (VSWR) Test

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production antennas shall be delivered prior to the approval of the preproduction model. Prefabrication of production antennas prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of the antennas to be submitted for acceptance. The preproduction model shall not be considered as one of the antennas under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following

(1) Individual Tests

(2) Sampling Tests

4.3.1 Individual Tests - Each antenna submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material and workmanship. As a minimum, each equipment accepted shall have passed the following tests

(1) Examination of Product

(2) Insulation Test

(3) Voltage Standing Wave Ratio (VSWR) Test

4.3.2 Sampling Tests - Sampling tests shall be conducted on the antennas and antenna base which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

| Quantity of Antennas and Antenna Bases Offered for Acceptance | Quantity to be Tested |
|---|---|
| First 10 | 1 |
| 11 to 100 | 2 |
| 101 to 200 | 1 |
| Over 200 | 1 for each additional 200 or fraction thereof |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:27Z
Check the source to verify that this is the current version before use.

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) All of the tests listed under 4.2.3, except environmental tests unless deemed necessary by the Government Inspector.

4.3.3 Equipment Failure - Should a failure occur during the sampling test, the following action shall be taken.

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Test Methods

4.4.1 Environmental Tests - The environmental test shall be conducted in accordance with Specification MIL-T-5422. The term "satisfactory operation" as used in Specification MIL-T-5422 shall consist of the insulation test of 4.4.3 and the measurement of voltage standing wave ratio (VSWR) at frequencies specified in 4.4.5(2). Variation in voltage standing wave ratio of more than 50% at any one frequency during the progress of the service condition tests shall be cause for rejection. Where necessary because of equipment limitations, voltage standing wave ratio measurements can be made before and immediately after individual environmental tests.

4.4.2 Mechanical Stress Test - The antenna shall be fastened securely in a fixture by means of mounting bolts. The use of AB-117( * )/AP antenna base for supporting the antenna in this test is optional. The antenna shall be loaded laterally when the blade is in a horizontal plane. A uniformly distributed load shall be applied in increments until a static load of 6.9 pounds per square inch has been attained. No yielding of any structural part nor impairment of electrical performance shall result. The Antenna Base AB-117( * )/AP when subjected to this test shall have an AT-929( * )/AP antenna installed and the base mounted in a jig which supports it by means of its regular mounting bolts.

4.4.3 Insulation Test - The resistance of the insulation between the stub and the sleeve of the completely assembled antenna shall be measured in accordance with the best commercial practice using a self-contained and self-powered insulation resistance test set having a 500-volt generator and a scale range of 0-1000 megohms with an additional scale marking for infinite resistance. The d-c resistance shall be not less than 50 megohms.

4.4.4 Radiation Test - The radiation test shall consist of connecting the antenna through a 50-foot length of RG-21( * )/U cable to a monitored signal source tuned to one frequency within the frequency range of the antenna and observing the relative level of energy radiated from the antenna as indicated on the output meter of an r-f receiving device tuned to the same frequency as the signal source. All conditions of tests shall be identical for each antenna tested. The relative level of radiation shall not vary more than .5 db from an average level determined from the approved preproduction models under identical conditions and methods of test.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:27Z
Check the source to verify that this is the current version before use.

4.4.5 Voltage Standing Wave Ratio (VSWR) Tests - The voltage standing wave ratio shall be determined in accordance with the best engineering practice when the antenna is mounted centrally on the convex surface of a semi-cylindrical ground plane of 3.5 feet radius and 10 feet length. The voltage standing wave ratio (VSWR) measured at or corrected to the antenna connector shall not be greater than 5:1 on a 50 ohm coaxial transmission line for the frequency range from 80 to 320 megacycles per second (mc/s) for the AT-929( * )/AP, and not greater than 3 1 from 300 to 1000 mc/s for the AT-500( * )/AP when similarly measured.

(1) On Preproduction Models - measurements shall be made at the following frequencies.

| | |
|---|---|
| AT-929( * )/AP | 60, 80, 100, 120, 140, 160, 180, 200, 220, 240, 260, 280, 300, 320, and 340 mc/s |
| AT-500( * )/AP | 280, 300, 350, 400, 450, 500, 550, 600, 650, 700, 750, 800, 850, 900, 950, 1000, and 1020 mc/s |

(2) On Inspection Models - measurements shall be made at the following frequencies.

| | |
|---|---|
| AT-929( * )/AP | 80, 120, 160, 200, 240, 280, and 320 mc/s |
| AT-500( * )/AP | 300, 400, 500, 600, 700, 800, 900, and 1000 mc/s |

4.4.6 Examination of Product - Each antenna shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.5 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.6 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Bureau of Aeronautics

4.7 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with SAR-300.

## 5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:27Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Use - The AT-500( * )/AP and AT-929( * )/AP antennas are intended for use with airborne receiving equipment in the frequency range from 80 to 1000 megacycles per second. A 50-ohm r-f transmission line such as RG-8( * )/U is required between the equipment and the antenna.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request shall be accompanied by complete supporting information.

6.3 The parentheses ( * ), when used in the type designation, shall be replaced either by a number or letter furnished by the Bureau of Aeronautics upon application by the contractor, or assignment of nomenclature in accordance with 3.2.10. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.4 Ordering Data - Procurement documents should specify the level of packaging and packing desired.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be rebarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:27Z
Check the source to verify that this is the current version before use.